

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

20th May 2023

Subject SA1 – Health and Care

Time allowed: 3 Hours 15 Minutes (14.45 - 18.00 Hours)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions inside the cover page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *The answers are expected to be India Specific application for the syllabus and corresponding core reading. However, substantially the core reading material is still taken from material supplied by Actuarial Education Company which is meant for UK Fellowship examination. The core reading also contains some material which is India Specific, mostly the IRDA regulation. In view of this, it should be noted that focal point of answers is expected to be India Specific application. However, if application specific to any other country is quoted in the answer the candidate should answer the question with reference to Indian environment.*
3. *Attempt all questions beginning your answer to each question on a separate sheet.*
4. *Mark allocations are shown in brackets.*
5. *Please check if you have received complete Question Paper and no page is missing. If so, kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately. You are not allowed to carry the question paper in any form with you.

- Q. 1)** You are working in a Health & Care insurance company which sells individual Critical Illness (CI), Income Protection (IP) and Immediate Impaired Annuity (IIA) products. The CI product is with fixed 10 years policy terms where level annual premiums are payable for 10 years. CI and IP products covers only the respective morbidity risks and don't provide any death benefit. About 50% of the morbidity risks of CI and IP portfolio are reinsured where reinsurance premiums (based on the attained age, gender, reinsurance cover etc.) are payable annually in advance.

The IIA is a single premium immediate annuity product for impaired lives only where annual annuities are payable in arrear during lifetime and the single premium is paid as death benefit on death.

None of the above products offer any surrender benefit and the policies terminate on death of the life assured after paying the death benefit, if applicable.

The prevailing regulatory regime requires prospective prudent valuation of liabilities and a consistent approach for valuation of assets. It allows credit for reinsurance in valuation but stipulates to disclose reinsurance items explicitly in the financial statements. Accordingly, the Gross premium, reinsurance premium, Change in Gross policy liabilities, Change in Net policy liability, Change in Reinsurance Credit etc. are shown explicitly in the Revenue A/C of the company. The reinsurance credit is defined as the excess expected reinsurance claims over expected reinsurance premium and the Net policy liability is the Gross policy liability less the Reinsurance Credit.

The pricing and best estimate morbidity assumptions are same as the respective reinsurance premium rates for IP & CI products. The prudent statutory valuation assumptions include 20% Margin for Adverse Deviation (MAD) added to best estimate assumptions for all the relevant assumptions.

- i)** State the main reasons why the company might have reinsured the CI and IP portfolios. (6)
- ii)** Suggest suitable asset classes for the company's investments backing its liabilities of IP portfolio with justification. You should also indicate, with reasons, the further information you would require to refine your suggested asset mix. (14)
- iii)** The CI policies are being sold for last 12 years where the level of business & experience remained similar throughout. At the last financial year end, the gross new business and renewal premiums are shown as INR 50 crores and 300 crores respectively in the annual Revenue A/C of the company. The corresponding reinsurance premium for the new business and renewal business are shown as INR 15 crores and 130 crores respectively.
 - a)** Describe the progression of "Gross policy liability", "Reinsurance Credit" and "Net policy liability" which will be used for preparation of the Revenue A/C for such a single CI policy over its policy term. (12)
 - b)** Provide the indicative ranges of amounts for the actual year-end Revenue A/C entries of "Gross increase in policy liability", "Change in Reinsurance Credit" and "Change in net policy liability" related to the whole CI portfolio of the company. (3)
- iv)** The pricing assumption of the impaired lives mortality for the IIA product is 250% of the standard annuitant's mortality table.

- a) Explain how the prudent valuation assumption of mortality will be derived mentioning the direction of MAD. Explain the factors which determine whether the valuation assumption will be more or less than 250% of the standard annuitant's mortality table. (10)
- b) Mention briefly the impact of a 50% quota share reinsurance arrangement for the death benefit cover where the proposed annual reinsurance premium is also 250% of the standard annuitant's mortality table. (5)
- [50]**

Q. 2)

- i) You are the Chief Actuary of a Large Health Company, where the Marketing Head came up with the proposal of offering a Group Income Protection fixed benefit product.
- The scheme is to be designed in a manner such that it will pay benefit only when the number of claims of the entire group insured under the scheme is above a specified % of the total number of employees covered.
 - The employer who participates in this scheme will register the number of employees they will wish to register and will pay a fixed premium per employee per year at the beginning of the year.
 - If an employee is absent from work on account of sickness for more than six consecutive days during the policy year a claim will be triggered and recorded.
 - Towards the end of the year proportion of claims recorded for all participating employees are calculated and if it's above a specified % of claims of the total employees covered, then all covered employees will get a fixed amount regardless of their health status.
- a) What are the possible benefits and risk of this scheme to a participating employer, please describe? (9)
- b) What are the possible risks of this scheme to the Insurance Company covering the contract? (9)
- c) What kind of control can be put in place by the insurance company while operating this scheme to tackle some of the risks highlighted in point (b) above? (7)
- ii) The marketing head of the company has come up with the proposal of taking the above scheme to be launched in overseas market.
- a) What is the additional risk to the insurance company for launching the above product to a new market, please describe? (7)
- b) Discuss the possible sources of data that could be available to price these products and what could be risk mitigants to reduce morbidity risks? (12)

- c) The finance director is concerned about this proposal and wants to understand how the product launch in overseas market can affect the liquidity and solvency position of the company?

(6)
[50]
